

Differentiation of Logarithmic Function

1. Simplify these expressions involving logarithms to the base e :
 - a. $e \ln e$
 - b. $\frac{1}{e} \ln \frac{1}{e}$
 - c. $3 \log_e e^2$
 - d. $\ln \sqrt{e}$
 - e. $e \ln e^3 + e \ln e$
 - f. $\log_e e + \log_e \frac{1}{e}$
 - g. $\ln e^e$
 - h. $\ln(\ln e^e)$
 - i. $\ln(\ln(\ln e^e))$
2. Solve:
 - a. $\ln(x^2 + 5x) = 2 \ln(x + 1)$
 - b. $\ln(7x - 12) = 2 \ln x$
3. Use the chain rule to differentiate:
 - a. $\ln(x^2 + 1)$
 - b. $\ln(x^2 + 3x + 2)$
 - c. $\ln(2 - x^2)$
 - d. $\log_e(1 + 2\sqrt{x})$
4. Use the logarithm laws to help differentiate:
 - a. $\ln 7x^2$
 - b. $\ln 5x^3$
 - c. $\ln \sqrt{x}$
 - d. $\ln \frac{3}{x}$
 - e. $\log_e \sqrt{2 + x}$
 - f. $\ln \left(\frac{1+x}{1-x} \right)$
 - g. $\ln \sqrt[3]{x+1}$
 - h. $\ln(x\sqrt{x+1})$
5. Use the change of base formula to express these to base e , then differentiate them:
 - a. $\log_2 x$
 - b. $\log_{10} x$
 - c. $\log_2 5x$
 - d. $5 \log_3 7x$
6. Differentiate these functions using the product rule:
 - a. $x \ln x$
 - b. $x \ln(2x + 1)$
 - c. $(2x + 1) \ln x$
 - d. $\sqrt{x} \ln x$
7. Find the equation of the tangent to $y = \ln x$ at the point where $x = e^2$
8. Find the equation of the normal to $y = \ln x$ at the point where $x = \frac{1}{e}$. What is its x -intercept?
9. Use the logarithm laws to simplify the following, then differentiate them:
 - a. $\ln \left(\frac{\sqrt{x}-1}{x^2+1} \right)$
 - b. $\ln \left((x^2 - 2x)\sqrt{x} \right)$
 - c. $\log_e \pi^x$
10. Differentiate the following, using the chain, product and quotient rules and the logarithm laws:
 - a. $x^2 \ln x$
 - b. $\frac{\ln x}{x}$
 - c. $(\ln x)^2$
 - d. $(\ln x)^4$
 - e. $\frac{1}{1+\ln x}$
 - f. $\sqrt{\ln x}$
 - g. $(2 \ln x - 3)^4$
 - h. $\frac{1}{\ln x}$
 - i. $\ln(\ln x)$
 - j. $\log_x 3$
 - k. $\frac{x}{\ln x}$
 - l. $\log_x 3^x$
11. Find the point(s) where the tangent to each of these curves is horizontal:
 - a. $y = x \ln x$
 - b. $y = \frac{1}{x} + \ln x$
 - c. $y = x^2 \ln \frac{1}{x}$

12. For what value of x does the tangent to $y = \log_{10} x$ have gradient 1?
13. Differentiate the following using any appropriate technique. Use the logarithm laws whenever possible.

- $\ln(2x^2 - 3x)$
- $\ln\left(\frac{x-1-2x^2}{5}\right)$
- $\ln(1 + \ln x)$
- $\ln\left(x^{\frac{1}{2}} + \ln x\right)$
- $(x^2 + 2x) \ln \sqrt{x-2}$
- $\ln\left(\frac{(x-3)^4 \sqrt{x}}{x+1}\right)$

14. a. Show that $y = \frac{x}{\ln x}$ is a solution of the equation $\frac{dy}{dx} = \left(\frac{y}{x}\right) - \left(\frac{y}{x}\right)^2$.
- b. Show that $y = \ln(\ln x)$ is a solution of the equation $x \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0$.

Answer:

- a. e ; b. $-\frac{1}{e}$; c. 6; d. $\frac{1}{2}$; e. $2e$; f. 0; g. e ; h. 1; i. 0
- a. $x = \frac{1}{3}$; b. $x = 3$ or 4
- a. $\frac{2x}{x^2+1}$; b. $\frac{2x+3}{x^2+3x+2}$; c. $-\frac{2x}{2-x^2}$; d. $\frac{1}{2x+\sqrt{x}}$
- a. $\frac{2}{x}$; b. $\frac{3}{x}$; c. $\frac{1}{2x}$; d. $-\frac{1}{x}$; e. $\frac{1}{4+2x}$; f. $\frac{1}{1+x} + \frac{1}{1-x}$; g. $\frac{1}{3x+3}$; h. $\frac{1}{x} + \frac{1}{2(x+1)}$
- a. $\frac{1}{x \ln 2}$; b. $\frac{1}{x \ln 10}$; c. $\frac{1}{x \ln 2}$; d. $\frac{5}{x \ln 3}$
- a. $1 + \ln x$; b. $\frac{2x}{2x+1} + \ln(2x+1)$; c. $\frac{2x+1}{x} + 2 \ln x$; d. $\frac{2+\ln x}{2\sqrt{x}}$
- $x - e^2y + e^2 = 0$
- $ex + e^2y + e^2 - 1 = 0, \left(\frac{1}{e} - e, 0\right)$
- a. $\frac{1}{2x-2} - \frac{2x}{x^2+1}$; b. $\frac{2x-2}{x^2-2x} + \frac{1}{2x}$; c. $\ln \pi$
- a. $x(1 + 2 \ln x)$; b. $\frac{1-\ln x}{x^2}$; c. $\frac{2 \ln x}{x}$; d. $\frac{4(\ln x)^3}{x}$; e. $\frac{-1}{x(x+\ln x)^2}$; f. $\frac{1}{2x\sqrt{\ln x}}$; g. $\frac{8}{x}(2 \ln x - 3)^3$; h. $\frac{-1}{x(\ln x)^2}$; i. $\frac{1}{x \ln x}$; j. $\frac{-\ln 3}{x(\ln x)^2}$; k. $\frac{\ln x - 1}{(\ln x)^2}$; l. $\frac{\ln 3(\ln x - 1)}{(\ln x)^2}$
- a. $\left(\frac{1}{e}, -\frac{1}{e}\right)$; b. $(1, 1)$; c. $\left(\frac{1}{\sqrt{e}}, \frac{1}{2e}\right)$
- $x = \frac{1}{\ln 10}$
- a. $\frac{4x-3}{2x^2-3x}$; b. $\frac{1-4x}{x-1-2x^2}$; c. $\frac{1}{x(1+\ln x)}$; d. $\frac{(2+\sqrt{x})}{2x(\sqrt{x}+\ln x)}$; e. $\frac{2x+x^2}{2(x-2)} + 2(x+1) \ln \sqrt{x-2}$; f. $\frac{3-12x-7x^2}{2x(3+2x-x^2)}$
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